

Passing networks and Poisson goals

How far can probability and network theory go in analysing a football team?

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Abstract

How can probability and network theory help analyse a football team, both in numbers and in words? I study the 2011 and 2015 UEFA Champions League finals, which FC Barcelona won 3–1 against Manchester United and Juventus. Each team’s passing is modelled as a weighted, directed network, with distances equal to the reciprocal of the number of passes. Centrality measures, a player-removal (“red card”) test and community detection show that Barcelona in 2011 were a single tightly connected block built around Xavi. His removal alone would cut the team’s passing efficiency by 14%, and 91% of the team’s betweenness belonged to three players. In 2015 Barcelona had become a two-sided team built around Alves and Messi on the right. Comparisons with 1,000 random networks per team, and a bootstrap of the pass counts, separate genuine structure from noise. A Poisson attack–defence model of goals is derived from first principles and tested on 371 Champions League knockout matches. On its own it forecasts no better than the season’s home, draw and away rates; adding home advantage, and fitting a Dixon–Coles model, both improve it. Finally, StatsBomb event data place players at their real positions and measure the quality of every chance (expected goals, xG). Barcelona created 1.93 xG to 0.30 in 2011, a performance that wins about 80% of the time. In 2015 the most central players in Barcelona’s passing network were also the main builders of its chances. The two branches of mathematics are most useful together: network theory describes how a team plays, and probability, applied to chances rather than goals, measures how well it played.

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1 Introduction

Football is the world’s most watched sport, and for a long time it was one of the least quantified. Its flow is continuous. Goals are rare. Much of what makes a team good, such as how it keeps the ball, moves it and creates chances, never shows up in simple counts like goals and assists. Companies such as Opta and StatsBomb now record every pass and shot, so a match can be treated as data.

Two branches of mathematics fit this data naturally. **Network theory** treats players as nodes and passes as weighted, directed edges. Measures developed for social networks, the web and transport systems can then describe a team’s structure: who the ball flows through, who is easy to reach, and which groups of players combine. **Probability** treats goals as rare, random events. A Poisson model then turns estimates of attacking and defensive strength into probabilities for every possible score.

I apply both to two matches that make a natural pair: the Champions League finals of 2011 (Barcelona 3–1 Manchester United, at Wembley) and 2015 (Juventus 1–3 Barcelona, in Berlin). Barcelona won both 3–1, but they played very differently. The research question is:

How can the knowledge of probability and network theory help quantitatively and qualitatively analyse a football team?

Section 2 describes the data. Section 3 sets out the network measures and Section 4 applies them. It tests whether the patterns could arise by chance and places players at their real positions on the pitch. Section 5 derives the Poisson model, and Section 6 tests it on 371 matches. Section 7 measures the quality of chances with expected goals, and Section 8 links the passing networks to the chances they produced.

2 Data

Passing tables (Opta). For each team in each final, Opta’s data give the number of completed passes from every starting player to every other starting player over the whole match. Substitutes are left out, so every network has the same eleven nodes. In these finals no substitution was made before the 69th minute. The four 11×11 tables are in Appendix A.

Event data (StatsBomb). StatsBomb’s free Open Data release [16] includes both finals. It records every on-ball event: over 1,200 passes in the 2011 final and almost 1,000 in 2015, each with a location, recipient and outcome, as well as every shot with its expected-goals value. I use it to check the Opta tables, to find players’ real positions and to measure the quality of the chances.

Results (openfootball). To test the probability model on more than two matches, I use every Champions League result from 2011/12 to 2023/24, taken from the public-domain openfootball dataset [14]. These are 13 seasons with 96 group-stage matches and 23–29 knockout matches each.

Do the sources agree? Table 1 compares the Opta tables with the same passes counted from StatsBomb’s events. The totals differ by at most nine passes. Across all 110 possible passing links, the correlation is at least 0.98 for three teams and 0.92 for Juventus. Two independent data providers therefore give almost the same networks, and every player ranking in this paper is the same whichever source is used.

Table 1: Opta passing tables against StatsBomb event data: completed passes between starters, and the correlation between the two sources over all 110 possible passing links.

Team	StatsBomb formation	Opta	StatsBomb	Correlation
Barça 2011	4-1-2-2-1	707	716	1.00
Man Utd 2011	4-2-2-2	262	256	0.98
Barça 2015	4-1-2-2-1	512	510	0.99
Juventus 2015	4-1-2-1-2	276	278	0.92

3 Network theory

3.1 Passing networks

A team’s passing in a match is represented by a **weighted directed graph** with the eleven starters as nodes. Its **adjacency matrix** A has entries

$$A_{xy} = \text{number of completed passes from player } x \text{ to player } y, \quad A_{xx} = 0.$$

The graph is directed because A_{xy} and A_{yx} usually differ: goalkeepers pass far more to centre-backs than centre-backs pass back. The weight of the edge $x \rightarrow y$ measures the *strength* of the link. The total number of passes a player makes, $L_x^{\text{out}} = \sum_y A_{xy}$, is their **out-strength**. Figure 1 draws the four networks, with players placed according to their team’s formation.

3.2 Strength and distance

Several measures rely on **shortest paths** (geodesics): the cheapest route for the ball from one player to another. For that we need the *length* of each edge. A link used 30 times should be *shorter* than a link used twice, because the ball travels along it easily. Following López Peña and Touchette [10], I use

$$d_{xy} = \frac{1}{A_{xy}} \quad (A_{xy} > 0),$$

and the length of a path is the sum of the lengths of its edges. The geodesic distance D_{xy} is the length of the shortest path from x to y .

This choice matters. Many software packages treat an edge weight as a length by default, and feeding them the number of passes directly turns the network upside down. Figure 2 gives an example. Player A passes to B 30 times, and passes to B via C only twice on each leg. With length equal to the number of passes, the direct route costs 30 and the detour costs $2 + 2 = 4$. The “shortest” path then avoids the busiest link in the team, and C is credited with holding the team together. With $d = 1/A$, the direct route costs $1/30 \approx 0.03$ against $1/2 + 1/2 = 1$ for the detour, which is the sensible answer.

3.3 Player measures

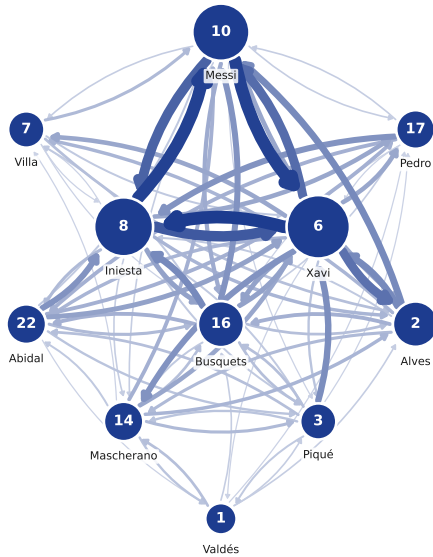
Betweenness centrality measures how much of the ball’s flow between other players depends on player x [8, 2]:

$$C_B(x) = \frac{1}{(N-1)(N-2)} \sum_{y \neq x \neq z} \frac{n_{yz}^x}{g_{yz}},$$

where g_{yz} is the number of geodesics from y to z and n_{yz}^x the number of those passing through x . With $N = 11$ the normalising factor is $1/90$, so $0 \leq C_B(x) \leq 1$. A high value means that isolating the player, by close marking or a red card, would disrupt the team’s passing.

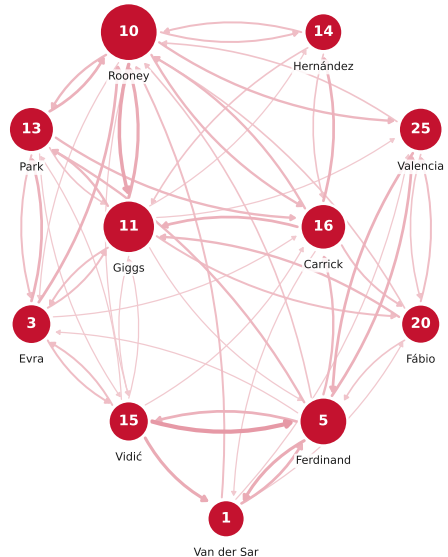
FC Barcelona passing network vs Man Utd

Champions League final 2011 · 4-3-3 · 707 passes between starters



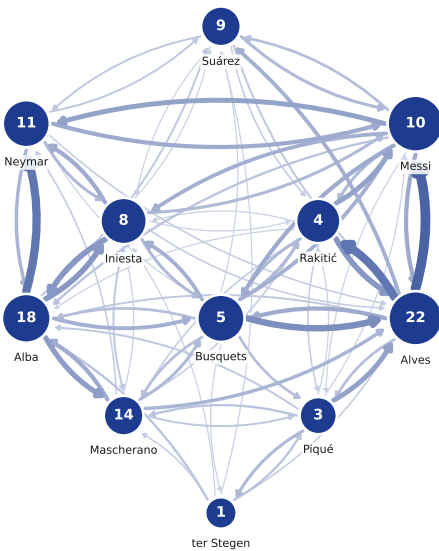
Man Utd passing network vs FC Barcelona

Champions League final 2011 · 4-4-2 · 262 passes between starters



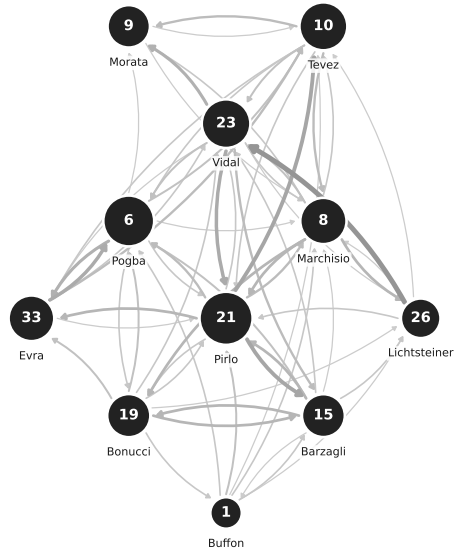
FC Barcelona passing network vs Juventus

Champions League final 2015 · 4-3-3 · 512 passes between starters



Juventus passing network vs FC Barcelona

Champions League final 2015 · 4-3-1-2 · 276 passes between starters



— 5 passes — 15 passes — 30 passes ● bigger circle = higher PageRank
Data: Opta. Arrow width is proportional to passes from one player to another; links with fewer than 2 passes are hidden. Goalkeeper at the bottom.

Figure 1: The four passing networks (Opta), with players placed by formation and the goalkeeper at the bottom. Arrow width is proportional to the number of passes, on the same scale for all four teams; circle size shows weighted PageRank (Section 3.3).

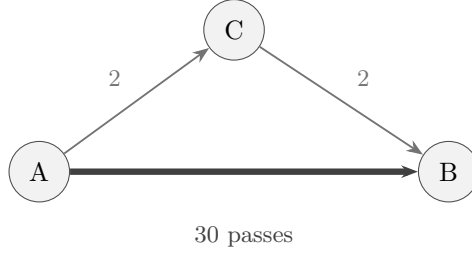


Figure 2: Why distance must be the reciprocal of the number of passes. With length = passes, the shortest route from A to B goes through C ($2 + 2 < 30$). With length = $1/\text{passes}$ it is the direct link ($1/30 < 1/2 + 1/2$).

Closeness centrality is the reciprocal of a player’s average geodesic distance to the others:

$$C_C(x) = \frac{N - 1}{\sum_{y \neq x} D_{xy}}.$$

Because distances are measured in units of $1/\text{passes}$, closeness is measured in passes. It is larger for teams that pass more, so it should only be compared between players in the same team.

PageRank [3] treats a player as important if important players pass to them often. The scores solve

$$x_i = \frac{1 - p}{N} + p \sum_j \frac{A_{ji}}{L_j^{\text{out}}} x_j, \quad \sum_i x_i = 1,$$

with $p = 0.85$. x_i can be read as the long-run share of time the ball spends with player i if it follows passes at random, with probability $1 - p$ of restarting from a random player.

Weighted clustering measures how often a player’s passing partners also pass to each other, forming triangles. For directed weighted graphs I use Fagiolo’s generalisation [7] of the formula of Onnela et al. [13]:

$$c_i^w = \frac{[(\hat{W}^{1/3} + (\hat{W}^T)^{1/3})^3]_{ii}}{2[k_i(k_i - 1) - 2k_i^{\leftrightarrow}]}, \quad \hat{W} = \frac{A}{\max A},$$

where the powers $1/3$ are taken entry by entry, k_i is the number of links (in either direction) between i and other players, and $k_i^{\leftrightarrow}$ the number of two-way links. Dividing by $\max A$ means only triangles of heavy links score highly.

3.4 Team measures

To describe a team as a whole I use the following measures.

- **Density:** the share of the $N(N - 1) = 110$ possible links used at least once.
- **Reciprocity:** $\sum_{x,y} \min(A_{xy}, A_{yx}) / \sum_{x,y} A_{xy}$, the share of passes that were returned along the same link.
- **Passing entropy:** $H = -\sum p_{xy} \ln p_{xy} / \ln 110$ with $p_{xy} = A_{xy} / \sum A$. It equals 1 if passes are spread evenly over all links and is lower if a few links dominate.
- **Betweenness centralisation** [8]: $\sum_x (\max_y C_B(y) - C_B(x)) / (N - 1)$. It equals 1 for a “star” in which every route runs through one player, and 0 when all players have equal betweenness.

- **Global efficiency** [9]: $E = \frac{1}{N(N-1)} \sum_{x \neq y} 1/D_{xy}$, the average ease of moving the ball between two players. Dividing E by the team’s average weight per link gives a *relative efficiency* that does not simply reward teams for passing more.
- **Modularity** [12]: how well the team splits into “passing units” that pass mainly among themselves (Section 4.4).

4 Results: the passing networks

4.1 Players

Table 2 lists the measures for every player. Three patterns stand out.

Barcelona 2011 ran through Xavi. Xavi has the highest betweenness (0.567), closeness, PageRank and clustering in his team. Only four Barcelona players have any betweenness at all: Xavi, Mascherano, Iniesta and Messi. By PageRank the top three are Xavi, Iniesta and Messi, the midfield triangle that dominates Figure 1.

Manchester United’s play ran from centre-back to striker. Ferdinand has the highest betweenness and closeness, and Rooney the highest PageRank. This is the route from defence to attack visible in Figure 1, with many of the passes going long. Giggs is the only midfielder close to them.

In 2015 Barcelona’s centre was on the right flank. Alves and Messi top both betweenness and PageRank. The right-back and the right-sided forward had become the hub of the team, where four years earlier it had been the central midfield. For Juventus, Pirlo has the highest betweenness, closeness and PageRank, followed by Marchisio and Vidal: their play ran through the centre.

4.2 Teams

Table 3 summarises each team in single numbers. Barcelona 2011 stands out. Its betweenness centralisation is 0.53, against about 0.19 for the other three teams, and 91% of its betweenness belongs to three players. It also has the most unequal involvement: Xavi was part of 39% of the team’s passes. Its low modularity says it played as one connected block rather than in separate units. It made more than 2.5 times as many passes as United.

So the Barcelona of 2011 was the *most* dependent of the four teams on a single player. That player was not a dribbler or a goalscorer, but a midfielder who touched the ball constantly. A team can look collective, with every player passing, while its structure still depends on one hub.

4.3 The red-card test

Betweenness only counts shortest paths. A more direct question is: *how much harder does it become for the other players to pass to each other if one player is removed?* For each player x , I compute the global efficiency among the other ten players with and without x available as an intermediary:

$$\Delta_x = 1 - \frac{E_{\text{without } x}(\text{others})}{E_{\text{with } x}(\text{others})}.$$

Averaging over the same pairs in both cases means $\Delta_x \geq 0$, and it measures only x ’s role as a link. The same calculation removes pairs and trios. Table 4 and Figure 3 show the results.

Removing Xavi costs Barcelona 2011 14% of its efficiency. No other player in that team is above 3%. The most damaging trio is Xavi, Busquets and Iniesta (31%). The midfield triangle of Xavi, Iniesta and Messi causes a 23% drop, the 9th largest of the 165 possible trios. For an opponent, the lesson is clear: stop Xavi, and the rest matters much less. For the other teams the dependence is spread more evenly. Barcelona 2015 relied most on the right-back Alves (7%), and Juventus relied equally on Pirlo and Vidal (4% each).

Table 2: Corrected player measures from the Opta passing tables, sorted by PageRank. C_B = normalised betweenness, C_C = closeness (in units of passes), PR = weighted PageRank ($p = 0.85$), c^w = weighted clustering.

Barça 2011					Man Utd 2011				
Player	C_B	C_C	PR	c^w	Player	C_B	C_C	PR	c^w
Xavi	0.567	9.59	0.178	0.227	Rooney	0.222	3.22	0.154	0.233
Iniesta	0.111	8.06	0.157	0.207	Giggs	0.106	2.81	0.130	0.206
Messi	0.089	7.84	0.149	0.214	Ferdinand	0.243	3.56	0.108	0.242
Busquets	0.000	6.95	0.097	0.179	Carrick	0.106	2.70	0.097	0.186
Alves	0.000	6.73	0.094	0.189	Park	0.006	2.65	0.095	0.222
Abidal	0.000	6.64	0.067	0.171	Valencia	0.067	2.64	0.088	0.185
Mascherano	0.183	6.50	0.066	0.137	Vidić	0.074	3.09	0.072	0.230
Pedro	0.000	5.82	0.062	0.146	Evra	0.026	3.10	0.070	0.247
Villa	0.000	3.95	0.052	0.116	Fábio	0.000	2.46	0.067	0.257
Piqué	0.000	7.31	0.051	0.120	Hernández	0.000	1.96	0.061	0.254
Valdés	0.000	3.33	0.028	0.085	Van der Sar	0.011	2.65	0.058	0.206

Barça 2015					Juventus 2015				
Player	C_B	C_C	PR	c^w	Player	C_B	C_C	PR	c^w
Messi	0.267	5.49	0.143	0.193	Pirlo	0.256	3.72	0.130	0.212
Alves	0.267	6.50	0.135	0.187	Pogba	0.028	2.40	0.120	0.172
Alba	0.111	5.56	0.110	0.174	Vidal	0.139	3.20	0.109	0.234
Busquets	0.144	6.43	0.103	0.191	Tevez	0.109	2.83	0.106	0.202
Neymar	0.011	4.28	0.102	0.170	Marchisio	0.163	3.52	0.098	0.189
Iniesta	0.067	4.68	0.098	0.143	Evra	0.011	2.34	0.095	0.203
Rakitić	0.000	4.91	0.089	0.156	Bonucci	0.083	2.76	0.084	0.166
Suárez	0.000	3.24	0.068	0.115	Barzagli	0.094	3.12	0.082	0.195
Mascherano	0.000	5.25	0.067	0.122	Morata	0.000	1.39	0.080	0.148
Piqué	0.144	5.72	0.056	0.118	Lichtsteiner	0.009	2.88	0.067	0.153
ter Stegen	0.000	3.52	0.029	0.105	Buffon	0.000	2.40	0.029	0.164

Table 3: Team-level network measures (Opta passing tables).

Measure	Barça 2011	Man Utd 2011	Barça 2015	Juventus 2015
Completed passes between starters	707	262	512	276
Density	0.85	0.75	0.82	0.78
Reciprocity	0.79	0.69	0.72	0.59
Passing entropy (normalised)	0.882	0.894	0.887	0.900
Betweenness centralisation	0.528	0.181	0.192	0.192
Betweenness held by top three	91%	66%	67%	62%
Involvement inequality (Gini)	0.311	0.175	0.211	0.162
PageRank inequality (Gini)	0.284	0.174	0.200	0.157
Mean weighted clustering	0.163	0.224	0.152	0.185
Relative efficiency	1.35	1.39	1.37	1.33
Modularity of passing units	0.056	0.156	0.152	0.093

Table 4: The red-card test: fall in passing efficiency between the remaining players when a player, pair or trio is removed.

Team	Most needed players	Most damaging pair	Most damaging trio
Barça 2011	Xavi (14%), Iniesta (3%)	Xavi + Iniesta (23%)	Xavi + Busquets + Iniesta (31%)
Man Utd 2011	Ferdinand (6%), Rooney (3%)	Ferdinand + Rooney (16%)	Ferdinand + Giggs + Rooney (28%)
Barça 2015	Alves (7%), Messi (4%)	Alba + Alves (15%)	Alba + Alves + Messi (25%)
Juventus 2015	Pirlo (4%), Vidal (4%)	Pirlo + Vidal (10%)	Bonucci + Pirlo + Vidal (18%)

4.4 Passing units

Community detection looks for groups of players who pass mainly among themselves. I combine passes in both directions and maximise the modularity

$$Q = \frac{1}{2m} \sum_{x,y} \left(w_{xy} - \frac{s_x s_y}{2m} \right) \delta(c_x, c_y),$$

where $w_{xy} = A_{xy} + A_{yx}$, s_x is player x 's total passes, $2m = \sum_{x,y} w_{xy}$ and $\delta(c_x, c_y) = 1$ if x and y are in the same group. The maximisation uses the Louvain algorithm [1], keeping the best of 20 runs. Table 5 lists the groups.

Barcelona 2011 has almost no structure ($Q = 0.056$). Six players (Alves, Busquets, Xavi, Iniesta, Messi and Villa) form a single unit. The centre-backs and goalkeeper form a second group, and Abidal and Pedro a small left-sided pair. Barcelona 2015 splits clearly into a right side (Alves, Busquets, Rakitić, Messi, Suárez) and a left side (Alba, Iniesta, Mascherano, Neymar). This is the clearest single measure of the change in style between the two finals. Manchester United divide into a left-sided and attacking group (Evra, Park, Giggs, Carrick, Rooney, Hernández) and a right-sided and defensive group (Van der Sar, the centre-backs, Fábio and Valencia). This fits both their long passes forward and Valencia's deep position (Section 4.6).

4.5 Could these patterns be chance?

A busy player will look central in almost any network, simply because they have more passes to spread around. To separate *who passed to whom* from *how much each player passed*, I compare

The red-card test: how much does each team need each player?

Efficiency = how easily the ball can travel between the remaining ten players, using 1/passes as distance.

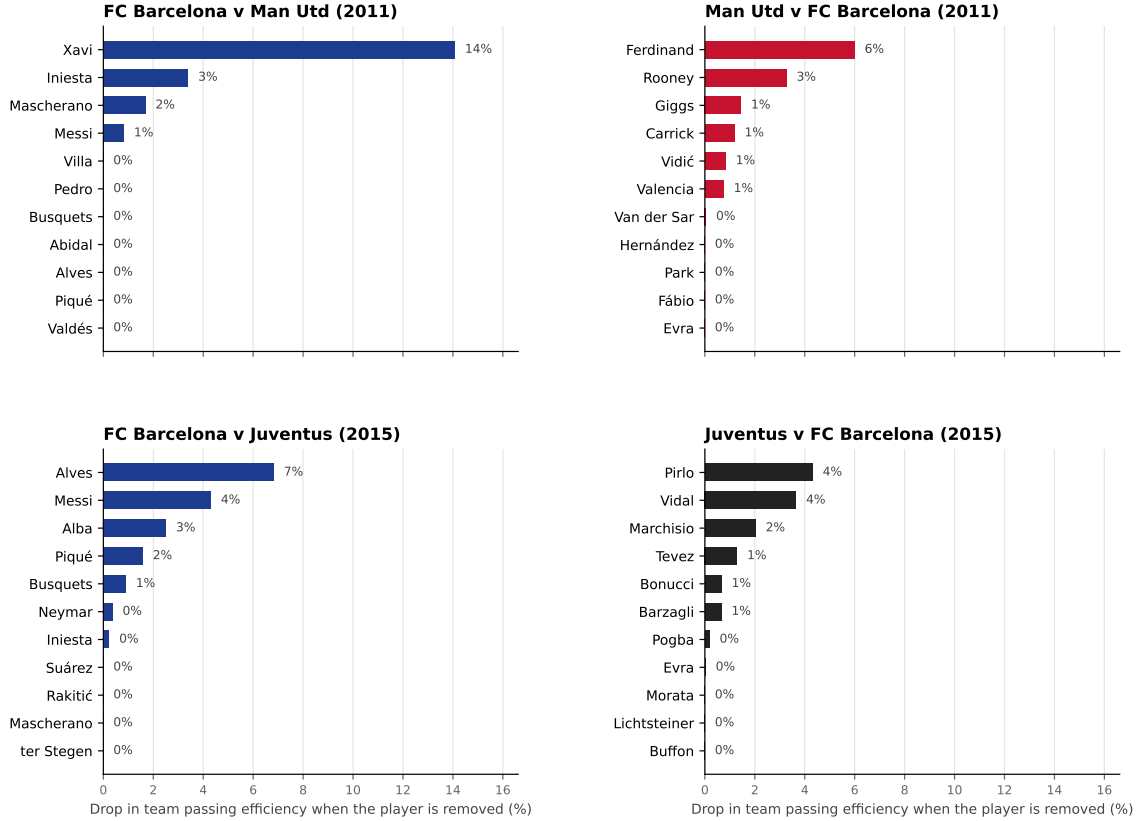


Figure 3: The red-card test for every player: the fall in passing efficiency between the remaining ten players when the player is removed.

each team with 1,000 random networks. In each one, every player makes and receives exactly as many passes as in the real match, but the receivers are shuffled at random, with no player passing to himself. If a real value lies far outside the random ones, it reflects genuine structure.

Table 6 gives the results as z -scores. Three findings are clear. First, **modularity** and **passing entropy** are far from random for every team: real teams concentrate their passes on particular links and form genuine units. Second, real teams are more **efficient** than random ones with the same pass totals. Third, **betweenness centralisation** is *not* unusual. Barcelona’s reliance on Xavi is fully explained by the number of passes he made and received, not by any special position in the network. The random comparison ignores the pitch, so a goalkeeper is as likely to pass to a striker as to a centre-back. It is a demanding baseline, and the clearest results should be read with that in mind.

How stable are the rankings? The pass counts are themselves noisy. A slightly different game could have produced a few more or a few fewer passes on each link. I resample every entry of A from a Poisson distribution with mean A_{xy} , 1,000 times, and record who comes out on top (Table 7). Xavi tops betweenness in 99% of resamples and the red-card test in 100%. In contrast, the order of Rooney and Ferdinand at United, and of Alves and Messi at Barcelona in 2015, changes often. These players are best described as “joint most important” rather than ranked.

Table 5: Passing units found by Louvain community detection (passes in both directions combined).

Team	Q	Units
Barça 2011	0.056	Alves, Busquets, Iniesta, Messi, Villa, Xavi; Mascherano, Piqué, Valdés; Abidal, Pedro
Man Utd 2011	0.156	Carrick, Evra, Giggs, Hernández, Park, Rooney; Ferdinand, Fábio, Valencia, Van der Sar, Vidić
Barça 2015	0.152	Alves, Busquets, Messi, Rakitić, Suárez; Alba, Iniesta, Mascherano, Neymar; Piqué, ter Stegen
Juventus 2015	0.093	Barzagli, Bonucci, Buffon, Pirlo; Lichtsteiner, Marchisio, Morata, Vidal; Evra, Pogba, Tevez

Table 6: Real networks compared with 1000 random networks per team that keep every player’s number of passes made and received. Entries are z -scores (standard deviations from the random mean); * marks $p < 0.05$ (two-sided).

Measure	Barça 2011	Man Utd 2011	Barça 2015	Juventus 2015
Reciprocity	−0.5	+0.8	−1.3	−1.9
Passing entropy	−13.8*	−7.2*	−19.5*	−6.4*
Betweenness centralisation	−0.4	−0.7	−0.9	−1.3
Top-three betweenness share	−2.7	−1.8	−2.8*	−2.2*
Mean weighted clustering	−0.4	−1.0	−4.0*	−2.6*
Relative efficiency	+4.5*	+5.9*	+11.3*	+4.3*
Modularity	+15.5*	+9.2*	+19.3*	+4.6*

4.6 Real positions

Placing players by formation is a simplification: footballers move constantly. StatsBomb’s event data let each player be placed at the average location of their passes and receptions (Figure 4). Several things become visible that a formation diagram hides:

- Barcelona 2011 squeezed into a compact block in the opponent’s half, with both full-backs level with the halfway line.
- Manchester United’s right winger Valencia played almost exactly level with right-back Fábio, in effect as a second full-back. StatsBomb records United’s shape as 4-2-2-2.
- Barcelona 2015 is visibly split into the left and right units found in Section 4.4, with Busquets between them.

Every player ranking in this section is unchanged when recalculated from the StatsBomb passes.

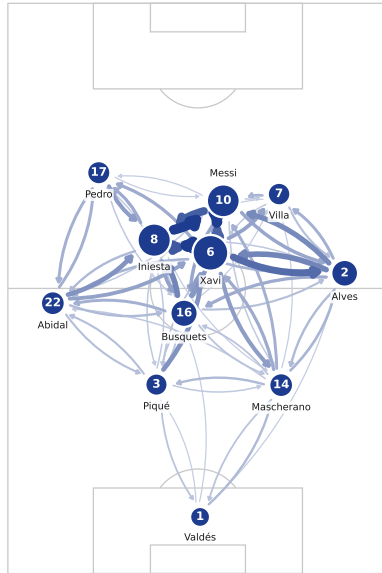
First half against second half. Splitting each match at half-time (Table 8, Figure 5) shows how the games changed. In 2015, Barcelona completed 336 passes between starters in the first half but only 174 in the second. Meanwhile the share of Juventus’s passes in the final third rose from 14% to 25%, as Juventus pushed for, and found, an equaliser. In 2011, Barcelona’s dependence on Xavi fell in the second half (centralisation 0.56 to 0.35) as Messi became more involved. Messi scored the decisive second goal nine minutes after half-time.

Passing networks at the players' real average positions

Starting XIs, completed passes between starters, whole match. Position = average location of the player's passes and receptions.

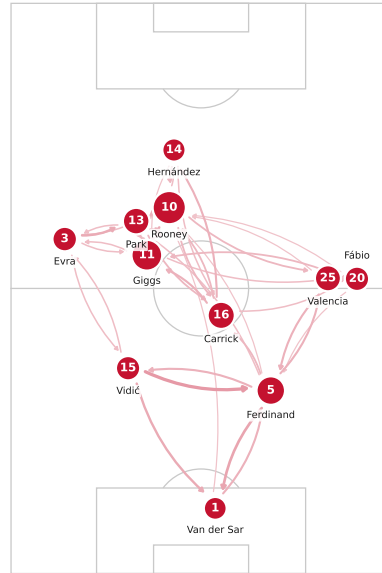
Barça 2011

716 completed passes · attacking ↑



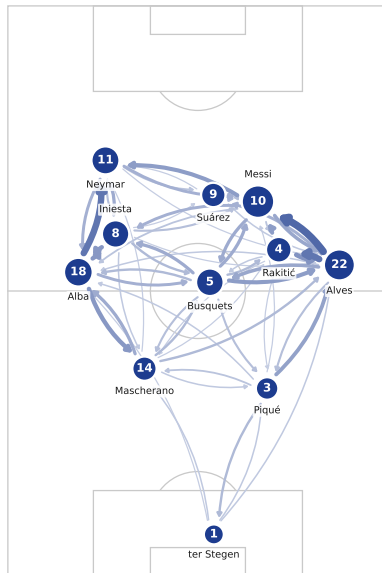
Man Utd 2011

256 completed passes · attacking ↑



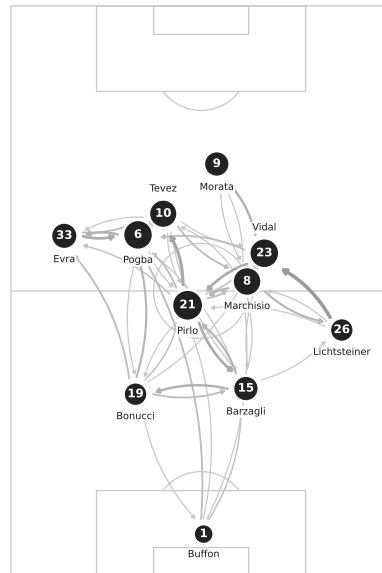
Barça 2015

510 completed passes · attacking ↑



Juventus 2015

278 completed passes · attacking ↑



— 5 passes — 15 passes — 30 passes ● bigger circle = higher PageRank

Data: StatsBomb Open Data. Links with fewer than 3 passes hidden. Overlapping circles nudged apart slightly.

Figure 4: Passing networks at the players' real average positions (StatsBomb Open Data), attacking upwards. Circle size shows PageRank; links with fewer than three passes are hidden. Data: StatsBomb.

Table 7: How often each player comes out on top when every pass count is resampled 1000 times from a Poisson distribution centred on the observed count (players topping fewer than 5% of resamples omitted).

Team	Top betweenness	Top PageRank	Top in red-card test
Barça 2011	Xavi 99%	Xavi 87%, Iniesta 11%	Xavi 100%
Man Utd 2011	Rooney 50%, Ferdinand 33%, Giggs 12%	Rooney 79%, Giggs 17%	Ferdinand 49%, Rooney 37%, Giggs 9%
Barça 2015	Alves 65%, Messi 22%, Busquets 6%, Alba 6%	Messi 68%, Alves 29%	Alves 76%, Messi 17%, Alba 5%
Juventus 2015	Pirlo 64%, Vidal 13%, Marchisio 12%	Pirlo 51%, Pogba 27%, Vidal 9%, Tevez 7%	Pirlo 59%, Vidal 25%, Marchisio 7%

Table 8: First half against second half (StatsBomb). Final-third share = proportion of a team’s completed passes that started in the attacking third.

Team	Half	Passes	Final third	Top PageRank	Betweenness centralisation
Barça 2011	1	387	24%	Xavi, Iniesta	0.559
	2	329	25%	Xavi, Messi	0.349
Man Utd 2011	1	139	29%	Giggs, Rooney	0.199
	2	117	17%	Rooney, Ferdinand	0.266
Barça 2015	1	336	24%	Alves, Messi	0.273
	2	174	28%	Messi, Alves	0.284
Juventus 2015	1	138	14%	Pirlo, Vidal	0.298
	2	140	25%	Tevez, Pogba	0.102

5 Probability: the Poisson model

5.1 From Bernoulli to Poisson

Divide a 90-minute match into n short intervals. Suppose that in each interval a team scores with a small probability p , independently of the other intervals, and cannot score twice in the same interval. Each interval is then a **Bernoulli trial**, with $P(1) = p$ and $P(0) = 1 - p$, or $f(k; p) = p^k(1 - p)^{1-k}$ for $k \in \{0, 1\}$. The number of goals X is a sum of n independent Bernoulli trials, so it follows the **binomial distribution**

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

For example, with $n = 6$ periods of 15 minutes, scoring in exactly periods 3 and 6 has probability $p^2(1 - p)^4$. There are $\binom{6}{2} = 15$ such arrangements, so $P(X = 2) = 15 p^2(1 - p)^4$.

The restriction of one goal per interval disappears as the intervals shrink. Let $n \rightarrow \infty$ while keeping the expected number of goals $\lambda = np$ fixed. Then

$$P(X = k) = \frac{n(n-1) \cdots (n-k+1)}{n^k} \cdot \frac{\lambda^k}{k!} \cdot \left(1 - \frac{\lambda}{n}\right)^n \cdot \left(1 - \frac{\lambda}{n}\right)^{-k} \rightarrow 1 \cdot \frac{\lambda^k}{k!} \cdot e^{-\lambda} \cdot 1,$$

since each factor $\frac{n-j}{n} \rightarrow 1$, $\left(1 - \frac{\lambda}{n}\right)^n \rightarrow e^{-\lambda}$ and $\left(1 - \frac{\lambda}{n}\right)^{-k} \rightarrow 1$. This is the **Poisson distribution**,

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots,$$

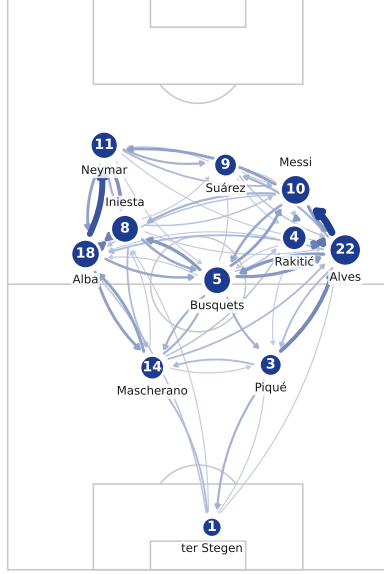
whose mean (and variance) is λ .

Barça 2015: first half vs second half

2015 final: Juventus 1–3 Barcelona

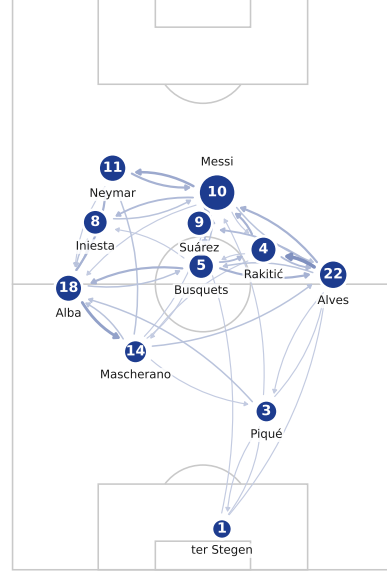
First half

336 completed passes between starters



Second half

174 completed passes between starters



— 3 passes — 8 passes — 15 passes • bigger circle = higher PageRank

Data: StatsBomb Open Data. Links with fewer than 2 passes hidden. Overlapping circles nudged apart slightly.

Figure 5: Barcelona 2015 by half (StatsBomb). The second-half network is much thinner as Juventus took more of the ball. Data: StatsBomb.

5.2 Attack and defence

Following Maher [11] and Spiegelhalter and Ng [15], the expected number of goals team i scores against team j is

$$\lambda_{ij} = \bar{g} \cdot \alpha_i \cdot \delta_j, \quad \alpha_i = \frac{\text{goals scored per game by } i}{\bar{g}}, \quad \delta_j = \frac{\text{goals conceded per game by } j}{\bar{g}},$$

where $\bar{g}$ is the average number of goals a team scores per game in the competition. α_i is team i 's *attack strength*: $\alpha_i = 1.4$ means 40% more goals than average. δ_j is team j 's *defensive weakness*. Substituting shows that λ_{ij} is simply (goals per game scored by i) $\times$ (goals per game conceded by j) $\div \bar{g}$. If the two teams' goals are independent Poisson variables with means λ_B and λ_O , then

$$P(\text{score } a-b) = \frac{\lambda_B^a e^{-\lambda_B}}{a!} \cdot \frac{\lambda_O^b e^{-\lambda_O}}{b!}, \quad P(\text{win}) = \sum_{a>b} P(a-b).$$

One league average. It might seem natural to use one average for goals scored and another for goals conceded. They are always the same. Every goal is scored by exactly one team and conceded by exactly one team, so over a whole competition

$$\sum_{\text{teams}} \text{goals scored} = \sum_{\text{teams}} \text{goals conceded}.$$

From the group stage onwards both seasons had 125 matches, with 355 goals in 2010/11 and 361 in 2014/15. That gives $\bar{g} = 355/250 = 1.42$ and $361/250 = 1.44$.

Table 9: The Poisson model for the two finals. $\bar{g}$ is the league average, λ_B Barcelona’s expected goals and λ_O the opponent’s. “Final included” estimates team strengths from all 13 games including the final itself; “final excluded” uses only the 12 games before it, which is the genuine forecast. Win, draw and loss are from Barcelona’s point of view, assuming independent Poisson scores.

Final	Version	$\bar{g}$	λ_B	λ_O	Win	Draw	Loss	$P(3-1)$
2011	Final included	1.420	0.875	0.713	37.0%	35.3%	27.7%	1.63%
2011	Final excluded	1.415	0.530	0.707	24.0%	41.0%	35.1%	0.51%
2015	Final included	1.444	1.270	0.766	48.3%	29.2%	22.5%	3.42%
2015	Final excluded	1.440	0.946	0.772	38.0%	33.6%	28.3%	1.95%

Only use information available before the match. A forecast must not contain the result it is forecasting. Each finalist played 13 games from the group stage onwards, the last being the final itself. Table 9 shows the model both ways. Estimated from 13 games, Barcelona look clear favourites in 2011 (37% against 28%). Estimated only from the 12 games before the final, the picture changes. United had conceded only 4 goals in those 12 games, a defensive weakness of $\delta = 0.24$, so Barcelona’s expected goals fall to 0.53, and the model makes United slight favourites (35% against 24%). In 2015 Barcelona remain favourites either way (38% against 28% before the final). Letting the result leak into the data made the model look better than it was.

The probability of the exact score 3–1 is small in every version: between 0.5% and 3.4%. That is not evidence against the model on its own. Even the most likely score (0–0 in the pre-match forecast, both years) has a probability of only 18–29%. A model has to be judged on many matches, which the next section does.

6 Testing the model

6.1 Method

I predicted all 371 knockout matches in 13 Champions League seasons (2011/12–2023/24). Each prediction used only the results of that season played before the match: the group stage plus any earlier knockout games. The outcome is the result after 90 minutes: 179 home wins, 72 draws and 120 away wins. Finals, and the single-venue ties of 2019/20, are treated as neutral. Five forecasts are compared:

1. **Guess:** 1/3 each for home win, draw and away win.
2. **Base rates:** the proportions of home wins, draws and away wins so far that season.
3. **The basic Poisson model** of Section 5.
4. **The basic Poisson model with home advantage:** $\bar{g}$ replaced by the average goals of home teams (for the home side) and of away teams (for the away side).
5. **A fitted Dixon–Coles model** [5]: $\log \lambda_{\text{home}} = \mu + h + a_i - d_j$ and $\log \lambda_{\text{away}} = \mu + a_j - d_i$, with attack a , defence d and home advantage h estimated by maximum likelihood. Because each team has only 6–12 earlier matches, the estimates are shrunk towards average by a normal prior with standard deviation 0.3. A correction factor τ adjusts the probabilities of 0–0, 1–0, 0–1 and 1–1, which plain Poisson models are known to get wrong:

$$\tau(0,0) = 1 - \lambda\mu\rho, \quad \tau(0,1) = 1 + \lambda\rho, \quad \tau(1,0) = 1 + \mu\rho, \quad \tau(1,1) = 1 - \rho,$$

where λ and μ here are the home and away expected goals and ρ is estimated with the other parameters.

Table 10: Forecasting 371 Champions League knockout matches (2011/12–2023/24), each from the same season’s earlier results. Lower is better for RPS, log loss and Brier score.

Model	RPS	Log loss	Brier	Favourite won
Guess 1/3 each	0.2454	1.099	0.667	–
Home/draw/away rates so far	0.2329	1.041	0.627	48.0%
Basic Poisson model	0.2467	1.106	0.659	46.6%
Basic Poisson + home advantage	0.2350	1.071	0.634	48.8%
Fitted Dixon–Coles model	0.2211	1.015	0.606	53.4%

Difference in mean RPS	Estimate	95% bootstrap interval
Basic Poisson – base rates	+0.0138	[−0.0062, +0.0350]
Basic Poisson + home – base rates	+0.0022	[−0.0158, +0.0213]
Dixon–Coles – base rates	−0.0118	[−0.0208, −0.0017]
Adding home advantage	−0.0116	[−0.0180, −0.0053]
Dixon–Coles – basic Poisson + home	−0.0139	[−0.0243, −0.0038]

Forecasts are scored with the **ranked probability score** [6, 4], the standard measure for football, together with log loss and the Brier score. For forecast probabilities (p_1, p_2, p_3) of home win, draw and away win, and the outcome (o_1, o_2, o_3) , one of which is 1,

$$\text{RPS} = \frac{1}{2} \sum_{k=1}^2 \left(\sum_{j \leq k} (p_j - o_j) \right)^2.$$

It is lower for better forecasts, and it rewards a forecast that is “close”: predicting a draw when the home team wins is less wrong than predicting an away win.

6.2 Results

Table 10 gives the results, with 95% bootstrap intervals for the differences.

- **The basic Poisson model is no better than the base rates.** Its mean RPS is slightly *worse*, although the interval includes zero. Knowing only how often home teams usually win beats it.
- **Home advantage is the biggest missing ingredient.** Adding it improves the basic model clearly (the interval excludes zero).
- **A fitted model does better still.** Dixon–Coles beats the base rates and the basic model with home advantage, with both intervals excluding zero. The favourite won 53% of the time under Dixon–Coles, against 47% for the basic model.

Figure 6 shows why the basic model struggles. It is **overconfident**. When it gave the home team about a 10% chance, home teams actually won a third of the time. When it predicted a draw at over 40%, fewer than one in ten of those matches were draws. Estimating team strengths from goals alone, from so few games, produces extreme values of α and δ , just as United’s $\delta = 0.24$ did in 2011. On average all three Poisson models predict more draws (24.0% for the basic model, 26.2% for Dixon–Coles) than the 19.4% observed. That fits the idea that teams in two-legged knockout ties play for a result over the two legs, not for a draw in one match.

Are the predicted probabilities right? (knockout matches 2011/12-2023/24)

Predictions grouped into bins of 0.2 (bins with at least 10 matches). Points on the dashed line mean the model's probabilities match reality.

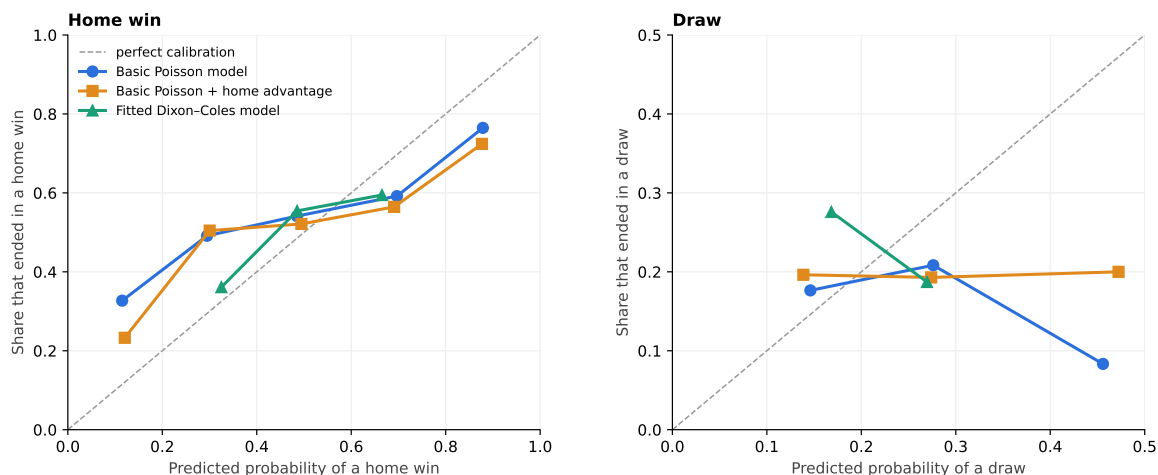


Figure 6: Calibration: predicted probability against how often the event happened, for home wins (left) and draws (right). A well-calibrated model lies on the dashed line.

Table 11: Shots and expected goals (Barcelona first). The simulated probabilities replay every chance 100,000 times; the last column is the pre-match Poisson forecast (final excluded, Table 9).

Final	Shots	xG	Goals	Barça win	Draw	Opp. win	$P(3-1)$	Pre-match Barça win
2011	22-4	1.93-0.30	3-1	80%	16%	4%	4.8%	24%
2015	17-15	2.27-1.38	3-1	59%	23%	18%	10.8%	38%

7 Expected goals: how good were the chances?

Goals are rare and noisy; chances are not. StatsBomb gives every shot an **expected-goals** value (xG): the probability that a shot from that position and situation is scored, estimated from many thousands of past shots. Adding a team's xG measures the quality of the chances it created, whatever the score.

To turn chances into result probabilities, I treat each attacking possession as a Bernoulli trial. Shots in the same possession, such as a shot and its rebound, are not independent. So a possession with shots of xG $q_1, \dots, q_m$ scores with probability $1 - \prod_k (1 - q_k)$. A team's goals are then a sum of independent Bernoulli trials with different probabilities, and I simulate that sum 100,000 times. This asks: *given the chances each team actually created, how often would this match end in each result?*

The two finals look very different (Table 11, Figure 7). In 2011, Barcelona out-shot United 22 to 4 and created 1.93 xG to 0.30. United's goal came from one of only four shots. Given those chances Barcelona win 80% of the time. Their dominance of the ball, which the networks show, turned directly into chances. The pre-match Poisson forecast, working only from goals in earlier games, gave Barcelona 24%.

The 2015 final was closer: 2.27 xG to 1.38, and a 59% chance of a Barcelona win. Rakitić scored in the fourth minute. Morata's equaliser came from a 0.60 xG chance early in the second half, during the spell when Juventus took over the ball (Table 8). Suárez restored the lead after 68 minutes, and Neymar scored in stoppage time. Here the pre-match forecast and the chances agree more closely (38% and 59%), but only xG shows how the match unfolded.

How good were the chances? Expected goals through each final

Each step is a shot, sized by its chance of going in. Dots mark goals. Data: StatsBomb Open Data.

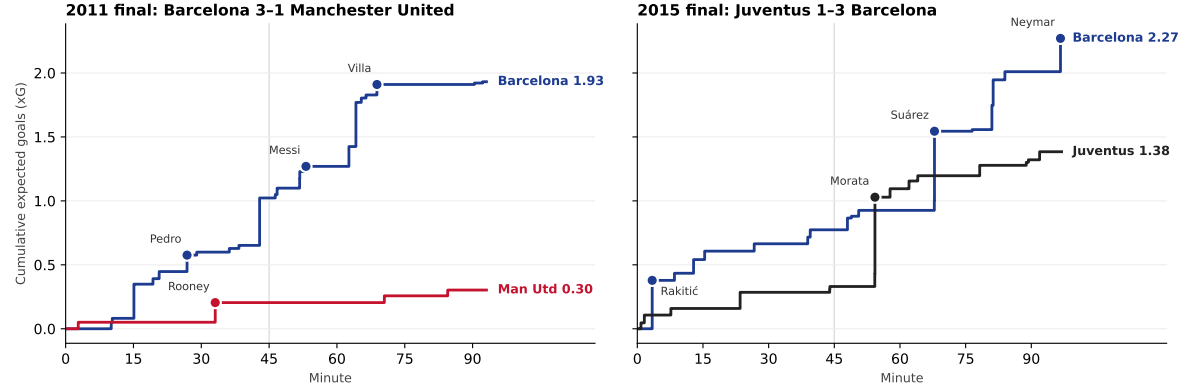


Figure 7: Cumulative expected goals through each final (StatsBomb). Each step is a shot; dots mark goals. Data: StatsBomb.

Table 12: Who was involved in the chances? xGChain and xGBuildup for the top three players in each team, and the Spearman rank correlation between PageRank (StatsBomb network) and xGBuildup across the eleven starters.

Team	Highest xGChain	Highest xGBuildup	ρ	p
Barça 2011	Messi 1.80, Busquets 1.65, Xavi 1.55	Busquets 1.57, Abidal 1.06, Xavi 1.04	0.31	0.36
Man Utd 2011	Rooney 0.30, Giggs 0.26, Carrick 0.26	Carrick 0.21, Giggs 0.20, Fábio 0.15	0.35	0.29
Barça 2015	Messi 2.02, Alves 1.66, Rakitić 1.59	Alves 1.43, Messi 1.33, Mascherano 1.08	0.65	0.03
Juventus 2015	Marchisio 0.95, Morata 0.94, Tevez 0.87	Marchisio 0.75, Lichtsteiner 0.20, Vidal 0.18	0.06	0.85

8 Linking the network to the chances

If passing networks describe how a team plays, they should say something about how it creates chances. Event data make this testable. For each player I compute:

- **xGChain:** the total xG of every shot-ending possession the player was involved in;
- **xGBuildup:** the same, but leaving out possessions in which the player took the shot or played the final pass. This measures contribution to the *build-up*.

If the passing network captures something real, the most central passers should also be the main builders of chances. Table 12 compares each player’s PageRank rank with their xGBuildup rank.

For Barcelona in 2015 the link is clear ($\rho = 0.65$, $p = 0.03$). Alves and Messi, the two most central players in the network, were also the two leading builders of chances. The right-sided play found in the network turned into chances. For Barcelona in 2011 the correlation is weaker (0.31). Busquets and Abidal contributed more to the build-up of chances than their network positions suggest, while Xavi, the network’s hub, is third. For Juventus there is no relationship. Almost all of their build-up passed through Marchisio, who was not their most central passer. With eleven players per team, these correlations are only suggestive, but they are a direct test of whether a team’s passing structure explains its attacking output.

9 Limitations

- **Two matches.** The network results describe two finals and four team performances. They illustrate what the methods can show; they do not establish general laws about Barcelona or about football.
- **Whole-match networks.** Aggregating over 90 minutes hides changes of tactics and game state. The half-by-half split is a first step; a finer split, for example into ten-minute windows or before and after each goal, would need more careful handling of small numbers.
- **Completed passes only.** Failed passes, carries and dribbles are not in the networks. Nor is the opponent: a pass under pressure counts the same as an easy one.
- **Starters only.** Substitutes are excluded. Because no substitution came before the 69th minute, the effect is small.
- **The random-network baseline ignores the pitch.** A goalkeeper is as likely to pass to a striker as to a centre-back. Any real team will look unusually structured against it. A stricter baseline would keep players' positions.
- **Small samples for the probability model.** Team strengths come from 6–12 earlier games. This is why the basic model is overconfident, and it limits any model based on goals. Shots or xG from earlier games would give more stable strengths, but they are not in the openfootball data.
- **xG is itself a model.** StatsBomb's xG values are estimates, and the simulation assumes that different possessions are independent.

10 Conclusion

How can the knowledge of probability and network theory help quantitatively and qualitatively analyse a football team? The answer has three parts.

Network theory describes *how* a team plays, provided the mathematics matches the football. When more passes mean a shorter distance, the measures agree with what the pictures show. They also quantify things the eye can only guess at. Barcelona in 2011 looked like a collective, but they were built around Xavi, whose removal would cost 14% of their passing efficiency, far more than any other player on either side. In 2015 they had become a two-sided team, with Alves and Messi at its hub. Random-network comparisons and resampling show which patterns are real (the passing units, the concentration of passes on particular links, Xavi's role) and which are fragile (the order of Rooney and Ferdinand, or of Alves and Messi).

A Poisson model of goals is a useful starting point, but on its own it is a weak forecaster. Tested on 371 matches with only the information available beforehand, the basic attack–defence model does no better than the season's home, draw and away rates. Adding home advantage and fitting the model properly both help, but goals are too rare for a handful of earlier matches to reveal a team's true strength.

The two approaches are strongest together, joined by richer data. Expected goals measure the quality of chances rather than the number of goals. They show that Barcelona's control of the ball in 2011 turned into a dominant set of chances, and that the 2015 final was closer than 3–1 suggests. Linking players' network positions to the chances they helped create tests whether passing structure explains attacking output. For Barcelona in 2015, it did.

Network theory and probability do not replace watching football. They make precise what a good observer suspects, show when that suspicion is wrong, and put a number on how much a result owed to skill and how much to chance.

Acknowledgements and reproducibility

All code is in the `recalc/` folder of the repository that accompanies this paper, and the tables are generated from it by `paper/make_tables.py`. `recalculate.py` computes the player measures and the Poisson model for the finals. `network_extra.py` contains the team measures, red-card test, passing units, random networks and bootstrap. `model_check.py` runs the forecasting test. `statsbomb_analysis.py` contains the event-data analysis; its data are downloaded by `fetch_statsbomb.py`. Random seeds are fixed, so rerunning the scripts reproduces every number. The analysis was carried out with the help of Claude, an AI assistant made by Anthropic.

Data. Passing tables: Opta. Event data: StatsBomb Open Data [16]; StatsBomb asks that any published work using its data credits StatsBomb as the source and displays its logo. Match results: openfootball (public domain) [14].

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A Passing tables

The four Opta passing tables used throughout.

Table 13: Completed passes between starters, Barça 2011 (Opta). Row = passer, column = receiver.

From / to	<i>Valdés</i>	<i>Piqué</i>	<i>Mascherano</i>	<i>Alves</i>	<i>Abidal</i>	<i>Xavi</i>	<i>Busquets</i>	<i>Iniesta</i>	<i>Pedro</i>	<i>Messi</i>	<i>Villa</i>
Valdés	0	3	6	3	0	1	3	1	2	0	1
Piqué	4	0	5	1	6	16	6	4	2	1	1
Mascherano	5	7	0	8	4	11	6	1	0	9	2
Alves	1	1	7	0	0	17	9	4	0	17	8
Abidal	2	5	0	0	0	12	7	15	9	4	1
Xavi	2	5	15	22	8	0	12	33	11	23	11
Busquets	0	6	4	6	9	12	0	17	3	15	4
Iniesta	0	4	3	8	7	27	12	0	11	32	2
Pedro	0	0	1	0	9	2	2	15	0	3	0
Messi	0	2	3	11	1	32	10	25	3	0	3
Villa	0	0	1	6	0	4	2	2	1	7	0

Table 14: Completed passes between starters, Man Utd 2011 (Opta). Row = passer, column = receiver.

From / to	<i>Van der Sar</i>	<i>Evra</i>	<i>Ferdinand</i>	<i>Vidić</i>	<i>Fábio</i>	<i>Giggs</i>	<i>Park</i>	<i>Valencia</i>	<i>Carrick</i>	<i>Rooney</i>	<i>Hernández</i>
Van der Sar	0	1	6	1	2	1	1	2	1	4	0
Evra	0	0	1	4	0	4	6	0	2	6	1
Ferdinand	7	2	0	6	0	1	5	6	5	3	0
Vidić	7	4	10	0	0	2	2	0	2	1	0
Fábio	0	0	3	0	0	5	0	4	0	3	0
Giggs	0	4	2	2	4	0	5	2	1	7	2
Park	0	4	0	2	1	3	0	1	5	6	1
Valencia	1	0	6	1	3	1	0	0	1	3	0
Carrick	2	1	1	1	4	6	1	1	0	5	6
Rooney	0	2	1	2	1	8	5	5	5	0	4
Hernández	0	0	0	0	0	3	1	1	3	4	0

Table 15: Completed passes between starters, Barça 2015 (Opta). Row = passer, column = receiver.

From / to	<i>ter Stegen</i>	<i>Piqué</i>	<i>Mascherano</i>	<i>Alba</i>	<i>Alves</i>	<i>Busquets</i>	<i>Iniesta</i>	<i>Rakitić</i>	<i>Messi</i>	<i>Neymar</i>	<i>Suárez</i>
ter Stegen	0	5	3	5	3	0	2	0	1	0	2
Piqué	7	0	5	4	12	0	1	2	2	0	1
Mascherano	1	4	0	10	8	7	2	3	1	3	1
Alba	1	0	13	0	1	9	14	0	0	21	4
Alves	0	6	1	4	0	10	1	21	25	0	9
Busquets	2	6	6	6	16	0	9	7	9	0	2
Iniesta	0	1	4	14	3	7	0	2	7	10	2
Rakitić	0	3	3	2	14	7	2	0	12	1	1
Messi	0	2	0	5	10	10	9	7	0	12	6
Neymar	0	0	1	8	3	2	7	1	10	0	4
Suárez	0	1	0	0	4	1	2	3	7	4	0

Table 16: Completed passes between starters, Juventus 2015 (Opta). Row = passer, column = receiver.

From / to	<i>Buffon</i>	<i>Barzagli</i>	<i>Bonucci</i>	<i>Lichtsteiner</i>	<i>Evra</i>	<i>Pogba</i>	<i>Marchisio</i>	<i>Pirlo</i>	<i>Vidal</i>	<i>Tevez</i>	<i>Morata</i>
Buffon	0	3	1	2	1	3	3	4	0	2	0
Barzagli	2	0	6	3	0	4	2	6	3	0	1
Bonucci	3	6	0	2	4	4	1	3	3	0	1
Lichtsteiner	1	1	0	0	0	1	2	3	12	2	1
Evra	0	0	1	0	0	8	1	2	3	5	0
Pogba	0	1	3	1	7	0	2	3	1	4	2
Marchisio	0	0	6	6	0	1	0	6	4	3	3
Pirlo	0	9	3	1	6	5	5	0	3	9	1
Vidal	0	5	0	2	2	4	3	8	0	0	7
Tevez	0	1	1	0	3	3	5	3	4	0	5
Morata	0	0	1	1	1	1	2	1	0	2	0